ANGEL CHEN

Interests: Data Science, Open & Reproducible Science

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EDUCATION

University of California, Santa Barbara (UCSB), Santa Barbara, CA — Sept 2017 – June 2021
Bachelor of Science in Statistics and Data Science

- High Honors

PROFESSIONAL EXPERIENCE

Data Scientist – Yale School of the Environment (YSE) – New Haven, CT — Nov 2024 – Present

- Advance the Malone Disturbance Ecology Lab (MDEL)'s projects by using version control to script reproducible workflows to clean, harmonize, analyze, model, and visualize diverse data sources.
- Provide technical support to 8 members of the MDEL by troubleshooting code and software.
- Utilize the Yale Bouchet high-performance computing (HPC) cluster and Slurm to build a data pipeline to download, compile, extract, and harmonize large datasets for downstream machine learning.
- Create metadata and perform quality control for 6 data packages for publication at the Environmental Data Initiative data repository.
- Design and deploy 2 interactive Shiny apps to visualize published data products from the MDEL.
- Collaborate with the Yale Center for Geospatial Solutions to create a public tool for estimating the fraction of fine particulate matter from biomass burning.
- Author the R packages "ERA5Flux" and "CarbonExchangeParameters".
- Create course materials and resources for YSE's Environmental Data Science in R: Introduction to Data Integration and Machine Learning (ENV 730).

Data Analyst – National Center for Ecological Analysis and Synthesis (NCEAS) – Santa Barbara, CA — Feb 2022 – Oct 2024

- Supported 12 Long Term Ecological Research (LTER) Network and 2 Delta Stewardship Council synthesis working groups by developing reproducible workflows to wrangle, analyze, visualize, and integrate various sources of data.
- Wrote content on the LTER SciComp Team website (https://lter.github.io/scicomp/), ranging from best practices for open science to tutorials on how to access certain data repositories and deploy R Shiny apps on a server.
- Promoted an open approach to synthesis science by hosting 11 workshops on collaborative coding with GitHub and the Tidyverse for the LTER synthesis working groups, LTER Information Managers, 2022 All Scientists' Meeting, and the 2023 American Geophysical Union conference.
- Maintained and contributed to the R package "scicomptools" on the Comprehensive R Archive Network (CRAN).
- Taught in 3 one-week-long courses on reproducible techniques and tools for research as part of the NCEAS Learning Hub Team.

Data Curator – Arctic Data Center – Santa Barbara, CA — Sept 2021 – Feb 2022
Data Curator Intern – Arctic Data Center – Santa Barbara, CA — Oct 2019 – Sept 2021

- Utilized R to build metadata and archive ecological data from research projects focused on environmental issues in the Arctic for the Arctic Data Center, a National Science Foundation data repository.
- Communicated with researchers about their file submissions for data quality assurance before publishing their finalized datasets with DOI numbers on the Arctic Data Center.

- Resolved over 100 tickets created by users each time they submitted datasets in an email ticketing system.
- Created metadata records for 7 datasets that informed key findings for the 2020 Arctic Report Card, published by the National Oceanic and Atmospheric Administration.

PUBLICATIONS

Maurer, G. E. et al. Ten simple rules for team synthesis in ecological research. *[In review at Methods in Ecology and Evolution]*

White, M. et al. Scale-dependent effects of species richness and asynchrony regulate the temporal stability of consumer-mediated nutrient dynamics. *[In review at Proceedings of the National Academy of Sciences]*

Malone, S. L. et al. A continental benchmark dataset for evaluating ecosystem gradient-flux approaches across 47 NEON flux towers. *[In prep]*

Macias, D. et al. Contrasting drivers of seed-production tempo at different scales of comparison. *[In prep]*

Ohlert, T. et al. (2025). Drought intensity and duration interact to magnify losses in primary productivity. *Science, 390*(6770), 284-289. doi.org/10.1126/science.ads8144.

Malone, S. L., McLeod, G., Chen, A. & Tupaj, M. (2025). Contemporary Fire Regimes of the Subtropical Everglades. *Sci Data, 12*, Article 1622. doi.org/10.1038/s41597-025-05917-6.

Brun, J. et al. (2025). Enabling data-driven collaborative and reproducible environmental synthesis science. *Methods in Ecology and Evolution, 16*(6), 1061–1074. doi.org/10.1111/2041-210X.70036.

Nigro, K. M. et al. (2025). Co-mast: Harmonized seed production data for woody plants across US long-term research sites. *Ecology, 106*(1), Article e4463. doi.org/10.1002/ecy.4463.

LaMontagne, J. M. et al. (2024). Community Synchrony in Seed Production is Associated With Trait Similarity and Climate Across North America. *Ecology Letters, 27*(12), Article e14498. doi.org/10.1111/ele.14498.

SELECTED DATA PRODUCTS

Malone, S. L., McLeod, G., Chen, A. & Tupaj, M. (2025). Fire in the Everglades, 1978-2023 ver 1. Environmental Data Initiative. doi.org/10.6073/pasta/72db80a33574c8ad936673d720c9360f.

White, M. et al. (2024). Estimates of nitrogen and phosphorus excretion rates in individual marine and estuarine animals ver 1. Environmental Data Initiative. doi.org/10.6073/pasta/379070279fa60341420f85157f66e338.

SOFTWARE & WEB APPS

Chen, A. et al. (2025). ERA5Flux: Improve the Utility of ERA5 Data at Flux Site Locations. R package version 0.0.0.9000. github.com/Malone-Disturbance-Ecology-Lab/ERA5Flux.

Chen, A. et al. (2025). CarbonExchangeParameters: Carbon Flux Partitioning Methods by the Malone Disturbance Ecology Lab. R package version 0.0.0.9000. github.com/Malone-Disturbance-Ecology-Lab/CarbonExchangeParameters.

Chen, A. (2025). Everglades National Park Fires. R Shiny App. angelchen.shinyapps.io/ENP_fires.

Chen, A. (2025). RustMapper: White Pine Blister Rust (WPBR) Risk. R Shiny App. angelchen.shinyapps.io/WPBR.

Chen, A. & Lyon, N. J. (2025). scicomptools: Tools Developed by the NCEAS Scientific Computing Support Team. R package version 1.1.1. doi.org/10.32614/CRAN.package.scicomptools.

Chen, A. & Lyon, N. J. (2024). ltertools: Tools Developed by the Long Term Ecological Research Community. R package version 1.0.0. doi.org/10.32614/CRAN.package.ltertools.

SELECTED COURSES, WORKSHOPS, & PRESENTATIONS

Downs, M. et al. (2024, Sept 16 – 20). Synthesis Skills for Early Career Researchers Reproducibility Best Practices. LTER Network Office, Santa Barbara, CA, United States. Course. lter.github.io/ssecr/mod_reproducibility.html.

Chen, A. & Lyon, N. J. (2024, Sept 5). Synthesis Skills for Early Career Researchers Version Control Module. LTER Network Office, Virtual. Course. lter.github.io/ssecr/mod_version-control.html.

Maurer, G. E. et al. (2024, Aug 4). Ecological Data Synthesis: A Primer of Essential Methods. Ecological Society of America Annual Meeting, Long Beach, CA, United States. Short course. lter.github.io/eco-data-synth-primer.

Chen, A. & Poulsen, C. V. (2024, June 10 – 13). NCEAS coreR for Delta Science Program. NCEAS Learning Hub, Sacramento, CA, United States. Course. learning.nceas.ucsb.edu/2024-06-delta.

Clark, J. et al. (2024, March 25 – 29). Scalable and Computationally Reproducible Approaches to Arctic Research. Arctic Data Center, Santa Barbara, CA, United States. Course. doi.org/10.18739/A2C53F33M.

Clark, J. et al. (2024, Feb 26 – March 1). Reproducible Approaches to Arctic Research Using R. Arctic Data Center & NCEAS Learning Hub, Virtual. Course. doi.org/10.18739/A2QF8JM4T.

Chen, A. & Lyon, N. J. (2023, Dec 10). Collaborative Coding with GitHub. American Geophysical Union Annual Conference, San Francisco, CA, United States. Workshop.

Chen, A. & Lyon, N. J. (2023, Nov 28). Collaborative Coding with GitHub. LTER Network Office, Virtual. Workshop. lter.github.io/workshop-github/.

LaMontagne, J. M. et al. (2022, Sept 20). Cross-site synthesis: Patterns & drivers of plant reproduction across LTER sites. LTER All Scientists' Meeting, Pacific Grove, CA, United States. Poster Presentation.

Chen, A. & Lyon, N. J. (2022, Sept 20). Reproducible & Collaborative Coding with Git. LTER All Scientists' Meeting, Pacific Grove, CA, United States. Workshop.

Lyon, N. J., Chen, A. & Brun, J. (2022, Aug 12). Coding in the Tidyverse. LTER Network Office, Virtual. Workshop. lter.github.io/workshop-tidyverse/.

Bachelder, N. R. et al. (2021, Aug 5). Leveraging Large Biological Interaction Data to Quantify Plant Specialization by Bees. Ecological Society of America Annual Meeting, Virtual. Poster Presentation. escholarship.org/uc/item/33b2t2bq.

AWARDS & ACHIEVEMENTS

Robert P. McIntosh Award – Ecological Society of America 2025

Central Coast Data Science Fellowship – UCSB – $5000 2020 - 2021

College of Letters & Science Honors Program – UCSB 2018 - 2020

Alumni Association Scholarship – UCSB – $2000 2018 - 2019